Name: _____

TOPIC TEST 1

Chapter 8 Continuous random samples and the normal distribution

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Eight multiple-choice questions (8 marks).
- Part B: Ten short-answer questions (32 marks).
- Part C: Two analysis questions (17 marks).

Part A

Multiple-choice questions

Eight multiple-choice questions
1 mark each: 8 marks
Circle the correct answer.

1 Which of the following could be probability density functions on the intervals given?

I $f(x) = 0.2$ on the interval $[5, 10]$

II $f(x) = \frac{x^2}{3}$ on the interval $[0, 3]$

III $f(x) = \begin{cases} 0.125(x - 2) \text{ for } 2 \leq x \leq 6 \\ -0.125(x - 10) \text{ for } 6 \leq x \leq 10 \end{cases}$

IV $f(x) = 2x - 6$ on the interval $[3, 4]$

A All of I–IV

B I and II only

C I, II and IV

D Only I

E I and IV only **[1 mark]**

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2 The probability density function of the continuous random variable, X, on the interval $[4, 20]$ is $\text{pdf}(x) = \dfrac{3(4-x)(x-24)}{3584}$. What is the cumulative probability function?

A $\dfrac{-x(x^2 - 42x + 288)}{3584}$

B $\dfrac{x(x^2 - 42x + 288)}{3584}$

C $\dfrac{-3(x^2 - 42x + 288)}{3584}$

D $\dfrac{-x(x^2 - 42x + 288)}{3584}$

E $\dfrac{-3x(x^2 - 42x + 288)}{3584}$ **[1 mark]**

3 The expected value of a uniform probability density function on the interval $[12, 38]$ is:

A 24

B 19

C 25

D $\frac{1}{19}$

E $\frac{1}{26}$ **[1 mark]**

4 The standard deviation of a uniform probability density function on the interval $[10, 30]$ is:

A 33.3

B 5.8

C 4.5

D 10

E 6.1 **[1 mark]**

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5 What is the area under the standard normal distribution for $-1.2 \leq Z \leq 1.5$?

A 0.0035

B 0.0483

C 0.1151

D 0.8181

E 0.8849 **[1 mark]**

6 What is the standard normal score for $x = 21$ for a random normal variable with mean of 25 and standard deviation of 5?

A 0.8

B -0.8

C 0.84

D -0.84

E -0.168 **[1 mark]**

7 X is a random normal variable with a mean of 35.2 and a standard deviation of 3.7. Find the probability that a random value of X lies between 33 and 36.

A 0.3095

B 0.2057

C 0.0673

D 0.1383

E 0.0323 **[1 mark]**

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8 What is the value of the quantile $t_{0.4}$ for the normal random variable, X, with mean 24.3 and standard deviation 7.2?

A 0.00045

B 24.05

C 22.50

D 225.0

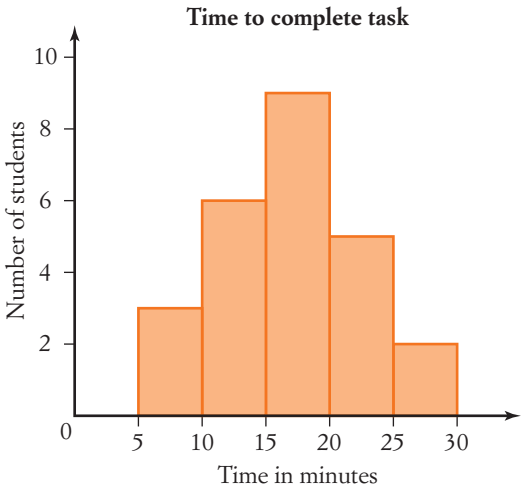
E 241.2 **[1 mark]**

Part B

Short-answer questions

Ten short-answer questions
32 marks
Show your working where appropriate.

1 The histogram below shows the times taken by some Year 12 students to complete a task. Use the histogram to find the probability that a randomly selected Year 12 student will complete the task within 12 minutes.



[3 marks]

2 Make a probability density function using $x^2 - 6x + 10$ on the interval $[2, 4]$.

[3 marks]

3 **a** Find the cumulative distribution function for the probability density function $f(X = x) = \frac{3}{32}(4x - x^2)$ defined on the interval $[0, 4]$.

b Use the cdf to find the probability $P(1 \leq X < 2.1)$.

[4 marks]

4 A rope thrown from a boat towards a jetty can land anywhere between the first and last mooring posts, that are 15 m apart. There are another two posts spaced equally between the first and last posts.

a Construct a probability density function for the distance the rope lands from the first post.

b Use the pdf to find the probability of the rope landing within a metre of a mooring post.

[3 marks]

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5 A measurement is given as 3.5 ± 0.2 m. Construct a triangular probability function for the measurement and find its mean and standard deviation.

[6 marks]

6 A continuous random variable, X, is transformed to the random variable, Y, according to the equation $Y = 3X + 4$. The mean and standard deviation of X are 13.4 and 3.5 respectively. What are the mean and standard deviation of Y?

[2 marks]

7 What is the probability density function of a normal distribution with $\mu = 21$ and $\sigma = 4$?

[2 marks]

8 Francis got 53 on a Maths test for which the class average was 39 and the standard deviation was 12. For another test, he got 59, but on this test the class average was 46 and the standard deviation was 9. In which test did he do better compared to the rest of the class?

[5 marks]

9 K is a standard normal variable. Find each of the following:

a $P(K < -0.2)$

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b $P(-0.2 \leq K \leq 1.3)$

[2 marks]

10 The mean of the times taken by Year 12 students to run 100 m was 18.3 seconds, with a standard deviation of 2.4 seconds. Assuming a normal distribution, what is the third quartile?

[2 marks]

Part C

Analysis questions

Two analysis questions
17 marks
Show your working where appropriate.

1 A pick-your-own strawberry farm monitors the mass of strawberries that customers pick. The most common mass is 1.8 kg, with a minimum of 0.4 kg and a maximum of 3.6 kg. Assuming that the distribution is triangular, what is the mean mass of strawberries picked?

[9 marks]

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2 In an art class, 30% of the students making a sketch complete it within 17 minutes, while 50% complete it within 20 minutes. Assuming a normal distribution, what percentage of the students complete it within 28 minutes?

[8 marks]

[Total marks: 57]

Answers

Part A – Multiple-choice questions [1 mark each]

1 Only I and IV have integrals of 1 $\therefore$ **E**

2 $\int \frac{3(4-x)(x-24)}{3584}dx$

$= \int \frac{-3x^2+84x-288}{3584}dx$

$= \frac{-x^3+42x^2-288x}{3584}$

$= \frac{-x(x^2-42x+288)}{3584}$

$\therefore$ **A**

3 $\frac{12+38}{2} = 25$

$\therefore$ **C**

4 $\frac{30-10}{2\sqrt{3}} \approx 5.8$

$\therefore$ **B**

5 Using a calculator, $P(-1.2 \leq Z \leq 1.5) \approx 0.8181$

$\therefore$ **D**

6 $Z = \frac{21-25}{5} = -0.8$

$\therefore$ **B**

7 Using a calculator, $P(33 \leq X \leq 36) \approx 0.3095$

$\therefore$ **A**

8 Using a calculator. invNorm(0.4, 24.3, 7.2) $\approx$ 22.50

$\therefore$ **C**

Part B – Short-answer questions

1 Total number of students = 25 **[1 mark]**

Number up to 12 minutes $= 3 + \frac{2.5}{5} \times 6 = 6$ **[1 mark]**

Probability of completion within 12 minutes $= \frac{6}{25} = 0.24$ **[1 mark]**

2 $\int_2^4 (x^2-6x+10)dx = \left[\frac{x^3}{3} - 3x^2 + 10x\right]_2^4 = \frac{8}{3}$ **[2 marks]**

pdf is $f(X=x) = \frac{3}{8}(x^2-6x+10)$ **[1 mark]**

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3 a $\text{cdf}(x) = \int_0^x \frac{3}{32}(4x - x^2)\,dx = \left[\frac{6x^2 - x^3}{32}\right]_0^x = \frac{6x^2 - x^3}{32}$ **[2 marks]**

b $P(1 \leq X < 2.1) = \text{cdf}(2.1) - \text{cdf}(1) \approx 0.3812$ **[2 marks]**

4 a $f(x) = \frac{1}{15}$ **[1 mark]**

b $P(\text{within 1 m}) = \int_0^1 \frac{1}{15}dx + \int_4^6 \frac{1}{15}dx + \int_9^{11} \frac{1}{15}dx + \int_{14}^{15} \frac{1}{15}dx = 0.4$ **[2 marks]**

5 The function will be 0.4 wide, so for the area to be 1, the height in the centre will be of height 5.
[1 mark for height]

Making each section a straight line gives:

$$f(x) = \begin{cases} 25(x - 2.8) & \text{for } 2.8 \leq x \leq 3 \\ -25(x - 3.2) & \text{for } 3 \leq x \leq 3.2 \end{cases}$$

[1 mark for slopes, 1 mark for each part of the equation]

mean = 3 (by symmetry) **[1 mark]** and SD $= \frac{3.2 - 2.8}{2\sqrt{6}} = \frac{1}{5\sqrt{6}} \approx 0.0816$ **[2 marks]**

6 $E(Y) = 3 \times 13.4 + 4 = 44.2$, $\text{SD}(Y) = 3 \times 3.5 = 10.5$ **[2 marks]**

7 $p(x) = \frac{1}{4\sqrt{2\pi}} e^{-\frac{(x - 21)^2}{2 \times 4^2}} = \frac{1}{4\sqrt{2\pi}} e^{-\frac{(x - 21)^2}{32}}$ **[2 marks]**

8 $Z_{\text{1st test}} = \frac{53 - 39}{12} = 1\frac{1}{6}$ **[2 marks]**

$Z_{\text{2nd test}} = \frac{59 - 46}{9} = 1\frac{4}{9} > 1\frac{1}{6}$ **[2 marks]**

Francis did better on the second test. **[1 mark]**

9 Using calculator:

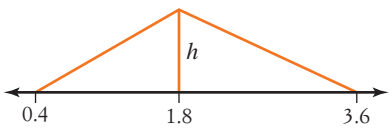
a $P(K < -0.2) \approx 0.4207$ **[1 mark]**

b $P(-0.2 \leq K \leq 1.3) \approx 0.4825$ **[1 mark]**

10 Using the inverse normal function with area = 0.75 on a calculator, Q3 ≈ 19.92 s **[2 marks]**

Part C – Analysis questions

1 The distribution is not symmetrical, so the areas either side of the maximum are not equal.



[1 mark]

Let the height at 1.8 be h.

Then $\frac{1}{2} \times 1.4 \times h + \frac{1}{2} \times 1.8 \times h = 1$

Thus $h = \frac{1}{1.6} = 0.625$. **[1 mark]**

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For $0.4 \leq x \leq 1.8, f(x) = \dfrac{0.625}{1.4}(x - 0.4) = \dfrac{0.625x}{1.4} - \dfrac{0.25}{1.4}$ **[1 mark]**

For $1.8 \leq x \leq 3.6\ f(x) = -\dfrac{0.625}{1.6}(x - 3.6) = -\dfrac{0.625x}{1.6} + \dfrac{2.25}{1.6}$ **[1 mark]**

$$\text{Mean} = \int_{0.4}^{3.6} f(x)x\,dx = \int_{0.4}^{1.8}\left(\frac{0.625x}{1.4} - \frac{0.25}{1.4}\right)x\,dx + \int_{1.8}^{3.6}\left(-\frac{0.625x}{1.6} + \frac{2.25}{1.6}\right)x\,dx$$

[2 marks for correct integral, 2 marks for correct evaluation]

$= 2.102\ldots.$ **[1 mark]**

The mean mass of strawberries is about 2.1 kg.

2 Since it is a normal distribution, $\mu = 20$ by symmetry. **[1 mark]**

For a standard normal distribution, the 0.3 quantile is at $Z = -0.5244\ldots$ from the inverse normal distribution function. **[2 marks]**

Thus $-0.5244\ldots = \dfrac{17 - 20}{\sigma}$, so $\sigma = 5.7208\ldots$ **[2 marks]**

Using the normal cdf function, $P(-\infty \leq 28) = 0.9190\ldots$ **[2 marks]**

About 92% will have completed the sketch within 28 minutes. **[1 mark]**

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